

DATA STRUCTURE LAB

DOUBLE LINKED LIST

The screenshot shows the SIMATS Online Programming Lab interface. The top header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", and the user's name "Nithiyasri S" with a dropdown arrow. Below the header, the left sidebar contains a "QUESTIONS" section with a dropdown menu set to "Single Linked List". The main area is titled "PROGRAM EDITOR" and contains a C code editor with the following code:

```
1 #include<stdio.h>
2 #include<stdlib.h>
3 struct Node
4 {
5     int data;
6     struct Node *next;
7 };
8 struct Node *head=NULL;
9 void insertatbegin(int value)
10 {
11     struct Node *newNode=malloc(sizeof(struct Node));
12     newNode->data=value;
13     newNode->next=head;
14     head=newNode;
15 }
16 void insertatend(int value)
17 {
18     struct Node *newNode=malloc(sizeof(struct Node));
19     newNode->data=value;
20     newNode->next=NULL;
21     if (head==NULL)
```

The right sidebar contains an "INPUT" section with the text "Your INPUT goes here!". Below the code editor, the "OUTPUT" section displays the result: "10 -> 15 -> 20 -> NULL".

The screenshot shows the SIMATS Online Programming Lab interface. The top header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", and the user's name "Nithiyasri S" with a dropdown arrow. Below the header, the left sidebar contains a "QUESTIONS" section with a dropdown menu set to "Deletion Operations". The main area is titled "PROGRAM EDITOR" and contains a C code editor with the following code:

```
1 #include<stdio.h>
2 struct Node
3 {
4     int data;
5     struct Node *next;
6 };
7 struct Node *head=NULL;
8 void display(){
9     struct Node *temp=head;
10    while(temp!=NULL){
11        printf("%d -> ",temp->data);
12        temp=temp->next;
13    }
14    printf("NULL");
15 }
16 void deletebeginning(){
17     struct Node *temp;
18     if(head==NULL)
19         return;
20     temp=head;
21     head=temp->next;
```

The right sidebar contains an "INPUT" section with the text "Your INPUT goes here!". Below the code editor, the "OUTPUT" section displays the result: "20 -> 30 -> 40 -> NULL".

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QUESTIONS 10 / 10 completed

Linked List

LINKED LIST

unplays its position is found.
Test Case 1

Input

List: 10 → 20 → 30 → 40
Search Element: 30

Output

Element found at position 3

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 #include<stdlib.h>
3 struct Node{
4     int data;
5     struct Node *next;
6 };
7 int main()
8 {
9     struct Node *head,*n1,*n2,*n3,*n4,*temp;
10    int key,pos=1,found = 0;
11    n1=malloc(sizeof(struct Node));
12    n2=malloc(sizeof(struct Node));
13    n3=malloc(sizeof(struct Node));
14    n4=malloc(sizeof(struct Node));
15    n1->data=10;
16    n2->data=20;
17    n3->data=30;
18    n4->data=40;
19    n1->next=n2;
20    n2->next=n3;
21    n3->next=n4;
```

OUTPUT

element found at position 3

INPUT

Your INPUT goes here!

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QUESTIONS 10 / 10 completed

Singly Linked List-f

SINGLY LINKED LIST-REVERSE

in reverse order write a C program to reverse a singly linked list and display the reversed list
Test Case 1

Input

10 → 20 → 30 → 40

Output

40 → 30 → 20 → 10 →

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 #include<stdlib.h>
3 struct Node{
4     int data;
5     struct Node*next;
6 };
7 int main(){
8     struct Node *n1,*n2,*n3,*n4;
9     int head;
10    n1=malloc(sizeof(struct Node));
11    n2=malloc(sizeof(struct Node));
12    n3=malloc(sizeof(struct Node));
13    n4=malloc(sizeof(struct Node));
14    n1->data=10;
15    n2->data=20;
16    n3->data=30;
17    n4->data=40;
18    n1->next=n2;
19    n2->next=n3;
20    n3->next=n4;
21    n4->next=NULL;
```

OUTPUT

INPUT

Your INPUT goes here!



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QUESTIONS 10 / 10 completed

Singly Linked List-1

SINGLY LINKED LIST-NODES

number of records currently stored. Write a C program to count and display the total number of nodes in a singly linked list.
Test Case 1

Input

10 → 20 → 30 → 40 → 50

Output

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node *next;
7 };
8
9 int countNodes(struct Node *head) {
10     int count = 0;
11     struct Node *temp = head;
12
13     while (temp != NULL) {
14         count++;
15         temp = temp->next;
16     }
17     return count;
18 }
19
20 int main() {
21     // Example Input: 10 -> 20 -> 30 -> 40 -> 50
```

OUTPUT

Total number of nodes = 5

INPUT

10->20->30->40->50->



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QUESTIONS 10 / 10 completed

Single Sorted Links

SINGLE SORTED LINKED LIST

Write preserving the sorted order. Write a C program to merge two sorted singly linked lists into a single sorted linked list.
Test Case 1

Input

List 1: 10 → 30 → 50
List 2: 20 → 40 → 60

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node* next;
7 };
8
9 struct Node* createNode(int value) {
10     struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
11     newNode->data = value;
12     newNode->next = NULL;
13     return newNode;
14 }
15
16 struct Node* mergeSortedLists(struct Node* a, struct Node* b) {
17     struct Node* mergedHead = NULL;
18     struct Node* mergedTail = NULL;
19
20     while (a != NULL && b != NULL) {
21         struct Node* temp;
```

OUTPUT

10 -> 20 -> 30 -> 40 -> 50 -> 60 -> NULL

INPUT

Your INPUT goes here!

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QUESTIONS 10 / 10 completed

Sorted Linked List-

SORTED LINKED LIST-2

a sorted linked list data displays the updated list.

Test Case 1

Input

10 → 10 → 20 → 20 → 30 → 30

Output

10 → 20 → 30 → NULL

PROGRAM EDITOR

C

RunSave

```
27 }
28
29 // Function to insert a node at the end
30 void append(struct Node** head_ref, int new_data) {
31     struct Node* new_node = (struct Node*)malloc(sizeof(struct Node));
32     struct Node* last = *head_ref;
33     new_node->data = new_data;
34     new_node->next = NULL;
35
36     if (*head_ref == NULL) {
37         *head_ref = new_node;
38         return;
39     }
40
41     while (last->next != NULL)
42         last = last->next;
43
44     last->next = new_node;
45 }
46
47 // Function to print the linked list
```

OUTPUT

INPUT

10->10->20->20->30->30

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QUESTIONS 10 / 10 completed

Circular Linked List

CIRCULAR LINKED LIST

write a C program to create a circular linked list and traverse all nodes exactly once.

Test Case 1

Input

10 → 20 → 30 → 40

Output

10 → 20 → 30 → 40 → Back to Head

PROGRAM EDITOR

C

RunSave

```
50 int main() {
51     struct Node* head = NULL;
52     int n, value;
53
54     // Note: For the specific test case format "10 -> 20 -> 30 -> 40"
55     // This implementation assumes you input the number of elements followed by the values
56     // If your lab environment uses a specific delimiter, you may need to adjust the scanf format
57
58     if (scanf("%d", &n) != 1) return 0;
59
60     for (int i = 0; i < n; i++) {
61         if (scanf("%d", &value) == 1) {
62             insertEnd(&head, value);
63         }
64     }
65
66     displayList(head);
67
68     return 0;
69 }
```

OUTPUT

INPUT

10->20->30->40->NULL



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QUESTIONS 10 / 10 completed

Doubly Linked List

linked lists are backward directions.

Test Case 1

Input

10 → 20 → 30 → 40

Output

Forward Traversal : 10 20 30 40

Backward Traversal: 40 30 20 10

PROGRAM EDITOR

C
Run Save

```

59     temp = temp->prev;
60 }
61 printf("\n");
62 }
63
64 int main() {
65     struct Node* head = NULL;
66     int n, value;
67
68     int inputs[] = {10, 20, 30, 40};
69     int numElements = 4;
70
71     for (int i = 0; i < numElements; i++) {
72         insertEnd(&head, inputs[i]);
73     }
74
75     displayForward(head);
76     displayBackward(head);
77
78     return 0;
79 }

```

OUTPUT

Forward Traversal : 10 20 30 40

INPUT

10->20->30->40



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QUESTIONS 10 / 10 completed

Polynomials

linked lists are perform polynomial addition.

Test Case 1

Input

Polynomial 1: 5x² + 3x + 2

Polynomial 2: 4x² + 6x + 1

Output

Polynomial 1: 5x² + 3x + 2

Polynomial 2: 4x² + 6x + 1

PROGRAM EDITOR

C
Run Save

```

1 #include <stdio.h>
2 #include <stdlib.h>
3
4 // Structure for a polynomial node
5 struct Node {
6     int coeff;
7     int exp;
8     struct Node* next;
9 };
10
11 // Function to create a new node
12 struct Node* createNode(int c, int e) {
13     struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
14     newNode->coeff = c;
15     newNode->exp = e;
16     newNode->next = NULL;
17     return newNode;
18 }
19
20 // Function to insert a term into the linked list in descending order of exponents
21 void insertTerm(struct Node** head, int c, int e) {

```

OUTPUT

Polynomial 1: 5x² + 3x + 2

Polynomial 2: 4x² + 6x + 1

INPUT

Poly 1: 5x² + 2
Poly 2: 4x² + 6x¹ + 1x⁰